

# Niklas Rieken

## Curriculum Vitae

Geschwister-Scholl-Straße 26

44135 Dortmund, Germany

\* November 9, 1992

+49 1523 8736069

rieken@oms.rwth-aachen.de

ttnick.github.io

in niklas-rieken



I am a computer scientist (BSc, MSc) with a PhD in Economics specializing in algorithmic methods, auction theory, and mathematical optimization. Having acquired a strong mathematical background with hands-on experience in Python, C/C++ and Linux-based workflows, I now seek for software-oriented roles where analytical rigor and algorithmic thinking are central.

### Core Skills

|             |                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------|
| Programming | <b>Python</b> (advanced), <b>Bash</b> (proficient), <b>PHP</b> (proficient), <b>C/C++</b> (intermediate) |
| Tools       | <b>git</b> (proficient), <b>Gurobi</b> (proficient), <b>LaTeX (+TikZ)</b> (advanced)                     |
| Systems     | <b>GNU/Linux</b> (proficient), <b>AWS</b> (basics, EC2, S3)                                              |
| Web & Data  | <b>HTML</b> (proficient), <b>CSS</b> (proficient), <b>SQL</b> (proficient)                               |
| Practices   | <b>agile Development</b> , <b>DevOps</b> , <b>Clean Code</b>                                             |
| Methods     | <b>Combinatorial Optimization</b> , <b>Graph Algorithms</b>                                              |
| Languages   | <b>German</b> (native), <b>English</b> (fluent), <b>Spanish</b> (basics)                                 |

### Work Experience

|           |                                                                                                                                                                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2019–2025 | <b>Doctoral Researcher</b> , <i>Chair of Management Science</i> , RWTH Aachen University<br>Researcher in Combinatorial Optimization and Algorithmic Game Theory.<br>Matchings, matroids, mechanism design, teaching courses in mathematical economics.        |
| 2017–2019 | <b>Research Assistant</b> , <i>Chair of Operations Research</i> , RWTH Aachen University<br>Developer for GCG ( <i>generic column generation</i> , MIP solver).<br>Performance tests, extracting statistics, and collecting information on vast instance sets. |
| 2014–2019 | <b>Teaching Assistant</b> , <i>Chair of Computer Science 7 / TCS</i> , RWTH Aachen University<br>TA for undergraduate courses in theoretical computer science.<br>Teaching exercise groups, grading assignments and exams.                                     |

### Education

|               |                                                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| November 2025 | <b>Doctorate (PhD)</b> , <i>Economics</i> , RWTH Aachen University<br>Thesis: Matroid Optimization in Auction Theory<br>Final Grade: summa cum laude (highest distinction)    |
| May 2019      | <b>Master of Science</b> , <i>Computer Science</i> , RWTH Aachen University<br>Thesis: Network Pricing Games with Atomic Splittable Followers<br>Final Grade: 1.4             |
| March 2017    | <b>Bachelor of Science</b> , <i>Computer Science</i> , RWTH Aachen University<br>Thesis: Alphabets and Machines as Structures and Their Decision Problems<br>Final Grade: 2.5 |

### Projects & Honors

|                |                                                                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Podcast        | <b>Algo2Go</b> , <i>topics in computer science and applied mathematics</i> , 2021<br>Co-hosted German podcast with Prof. Dr. Laura Vargas Koch and Dr. Björn Tauer to explain algorithms to a broad audience. Responsible for content planning, research, and production. |
| Teaching Award | <b>Prize for outstanding teaching activities</b> , 2024<br>Faculty 8 of Economics, RWTH Aachen University                                                                                                                                                                 |